



Hot Water Treatment and Postharvest Loss in Carabao Mango

Treatment × season effects on physiological and pathological loss

Synthetic data generated by an LLM for training purposes. Not for real-world decisions.

Total Batches

48

Starting Weight

15,279 kg

Pathological Loss

977 kg

Physiological Loss

1,005 kg

Est. Gross Savings

₱17,079

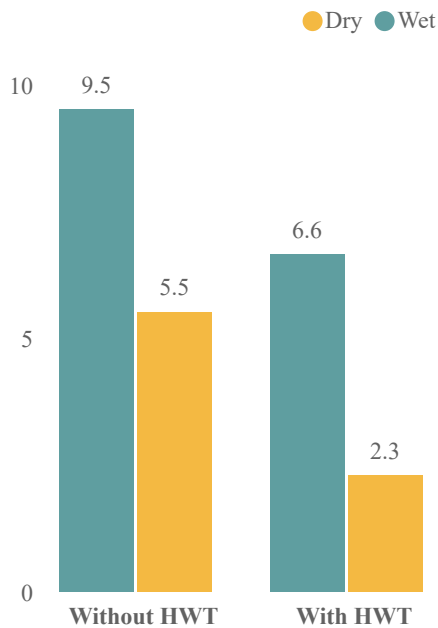
HWT Adoption Rate

33%

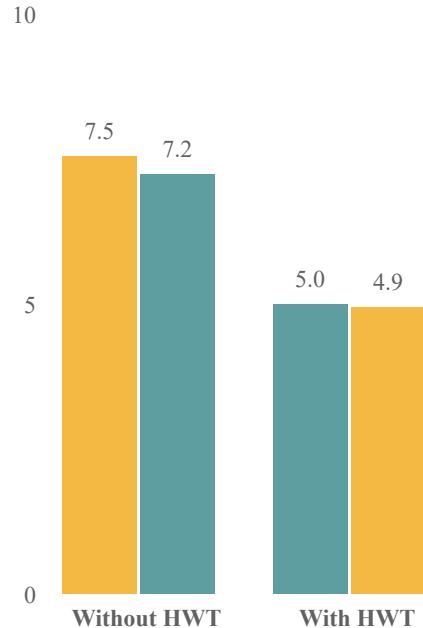
Money KPI Status

✓ Peso figures valid for this view.

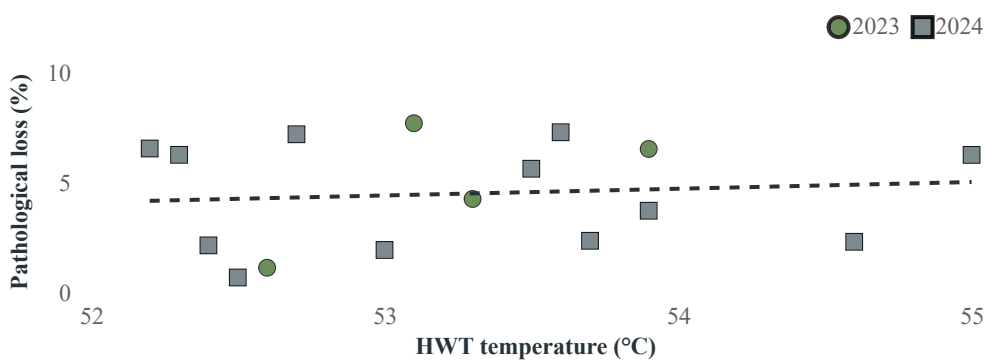
Pathological Loss (% per batch)



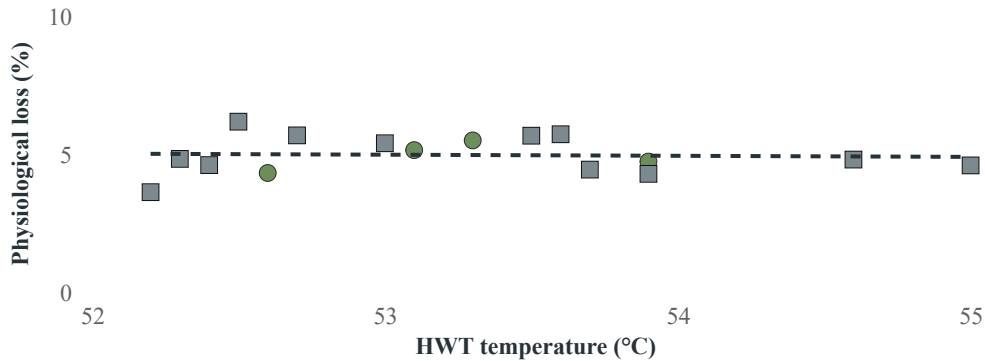
Physiological Loss (% per batch)



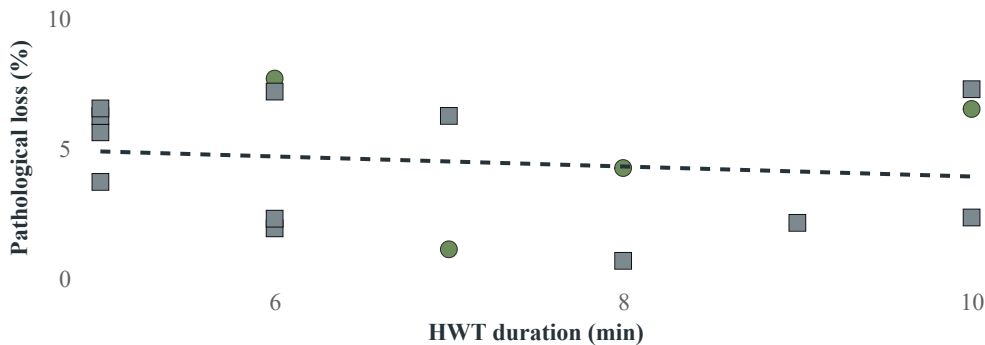
Pathological Loss vs. HWT Temperature



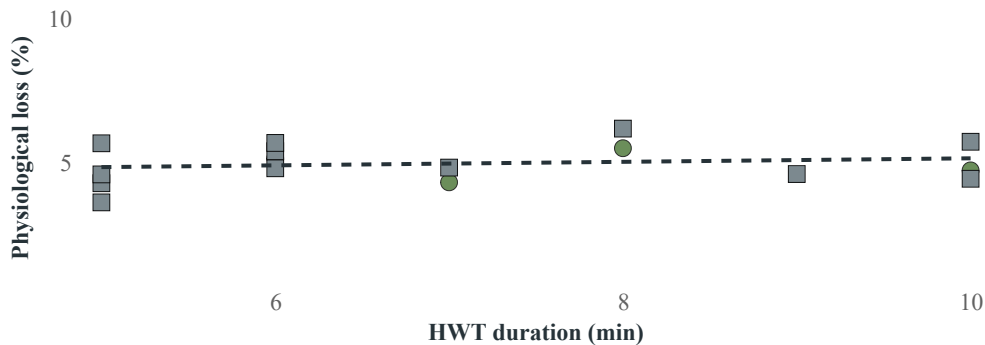
Physiological Loss vs. HWT Temperature



Pathological Loss vs. HWT Duration

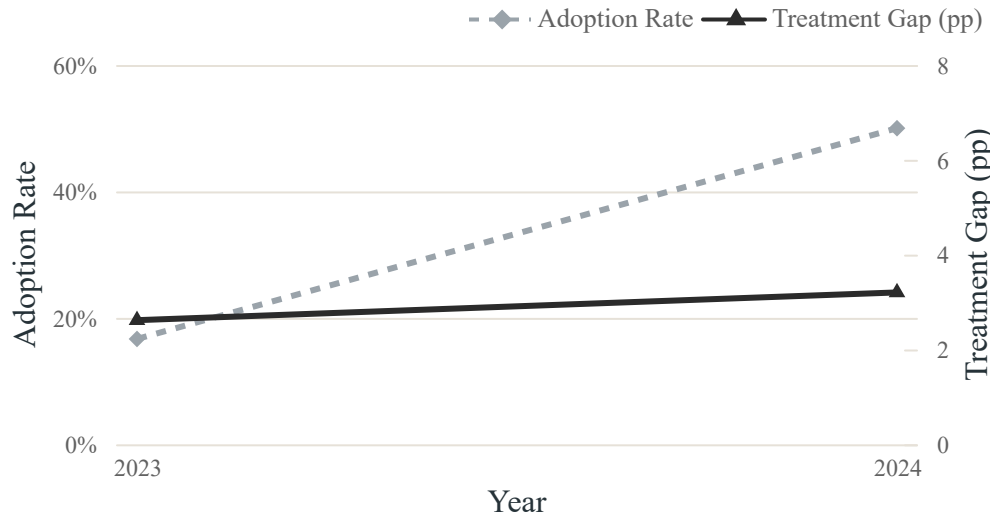


Physiological Loss vs. HWT Duration



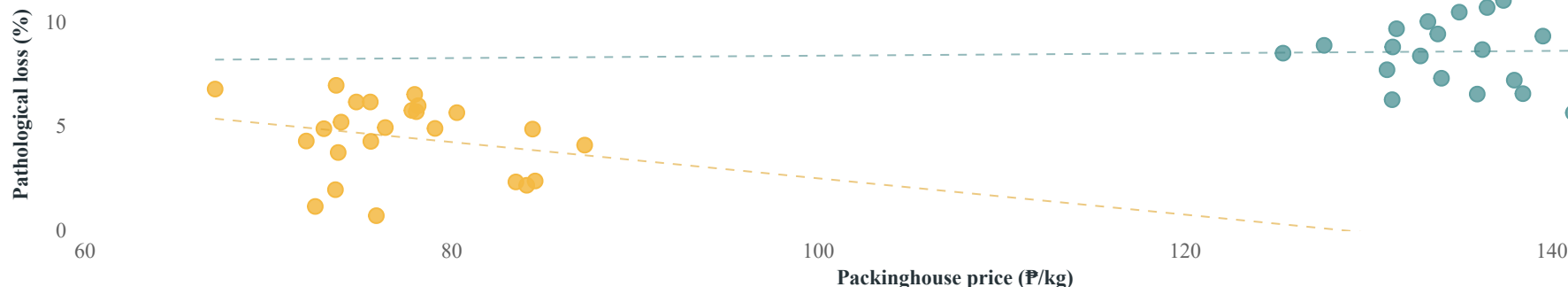
More Batches Treated, Same Effect per Batch

Adoption: 17% → 50% · Treatment gap: ~3 pp in both years



Price and Decay Move Together Only Because of Season

The overall trend comes from wet-season fruit being both pricier and more decay-prone



Season

All

Hot Water Treated

All

Year

All

Province

All

Type of Loss

Pathological Loss

Key Insights

- HWT results in the reduction of pathological and physiological loss by ~3 and ~2.4 percentage points (pp) per batch, respectively (~40% and ~33% relative reduction).
- Season has a significant effect on pathological loss, raising it by ~4 pp, but barely changes physiological loss (< 0.3 pp).
 - Wet season is consistent with increased risk of anthracnose and stem-end rot infection and proliferation under humid conditions.
- The increase in adoption rate (17% → 50%) reduces the loss in general, but the per-batch treatment effect does not change between years.

Recommendations

- Promote the use of HWT to further reduce loss, especially in wet season (Jun–Nov).
 - Pair HWT with additional decay control for wet-season fruit, since treated wet-season batches (6.6%) still lose more than untreated dry-season batches (5.5%).

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